



PT. SALIPI NAGARI PERKASA

Komp. Pejaten Mas Blok E No. 3 Kramat Watu Serang - Banten 42161

Telp. : 0254 - 231363 Fax.: 0254 - 231183

LAPORAN UJI MINYAK TRAFO

di

UNIT JASA PEMBANGKIT

JERANJANG



Yang dilakukan pada tanggal 07 Desember 2017

Untuk keadaan trafo bulan DESEMBER 2017

<u>1.0 Table of Contents</u>	<u>Page</u>
1.0 Specific Information	2
2.0 Our Comments and Recommendations of the Test Results Analysis	
- Transformer Unit MAIN TRAFO UNIT 1 (S/N: 1006901) Test Certificate 4	3
- Transformer Unit TRAFO 1 ESP UNIT 1 (S/N: 10937) Test Certificate 5	5
- Transformer Unit TRAFO 2 ESP UNIT 1 (S/N: 10035) Test Certificate 6	7
- Transformer Unit TRAFO 3 ESP UNIT 1 (S/N: 10031) Test Certificate 7	9
- Transformer Unit TRAFO 4 ESP UNIT 1 (S/N: 10034) Test Certificate 8	11
- Transformer Unit MAIN TRAFO UNIT 3 (S/N: 1122170) Test Certificate 9	13
- Transformer Unit UAT 3 (S/N: 1035082) Test Certificate 10	14
- Transformer Unit TRAFO 1 ESP UNIT 3 (S/N: 007308428) Test Certificate 11	15
- Transformer Unit TRAFO 2 ESP UNIT 3 (S/N: 0070084208) Test Certificate 12	17
- Transformer Unit TRAFO 3 ESP UNIT 3 (S/N: 0072084208) Test Certificate 13	19
- Transformer Unit TRAFO 4 UNIT 3 (S/N: 0071084208) Test Certificate 14	21
- Transformer Unit ASH HANDLING (S/N: 10820108) Test Certificate 15	23
- Transformer Unit ASH HANDLING (S/N: 10820107) Test Certificate 16	24
3.0 Summary of Recommendation	25

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	1006901	4	OLTT+ and Corrosive Sulfur
2.	TRAFO 1 ESP UNIT 1	10937	5	OLTT+ and Corrosive Sulfur
3.	TRAFO 2 ESP UNIT 1	10035	6	OLTT+ and Corrosive Sulfur
4.	TRAFO 3 ESP UNIT 1	10031	7	OLTT+ and Corrosive Sulfur
5.	TRAFO 4 ESP UNIT 1	10034	8	OLTT+ and Corrosive Sulfur
6.	MAIN TRAFO UNIT 3	1122170	9	OLTT+ and Corrosive Sulfur
7.	UAT 3	1035082	10	OLTT+ and Corrosive Sulfur
8.	TRAFO 1 ESP UNIT 3	007308428	11	OLTT+ and Corrosive Sulfur
9.	TRAFO 2 ESP UNIT 3	0070084208	12	OLTT+ and Corrosive Sulfur
10.	TRAFO 3 ESP UNIT 3	0072084208	13	OLTT+ and Corrosive Sulfur
11.	TRAFO 4 UNIT 3	0071084208	14	OLTT+ and Corrosive Sulfur
12.	ASH HANDLING	10820108	15	OLTT+ and Corrosive Sulfur
13.	ASH HANDLING	10820107	16	OLTT+ and Corrosive Sulfur

Note:

* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAF0 UNIT 1	1006901	4

A) Analysis

- **Acidity** of 0.011mgKOH/g and **IFT** of 40.2mN/m are **ACCEPTABLE**.
- **Dielectric** of 87kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** shows increasing in volume for Hydrogen, Carbon Monoxide and Carbon Dioxide. High Hydrogen and Carbon Oxides in a transformer's oil is an indication that the transformer is afflicted with a partial discharge condition. It is advisable to consult transformer manufacturer to ascertain the cause of fault gases buildup in the transformer.
- **Water Content** in oil of 7ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.46% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.27% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- Retest for **DGA** in 6 months to diagnose any possible fault and establish trend of the gases.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 1

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		45	NORMAL		GOOD	
31/07/2017	ONLINE	28	28				
30/08/2017	OFFLINE	49	49	NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	48		NORMAL	FAIR	GOOD	NONE
30/10/2017	ONLINE	60	64	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE		51				
06/12/2017	OFFLINE	50	48				

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.017 AC	41.5 AC	58 AC	0.5 AC	0.8691 AC	CLEAR AC	
31/07/2017	ROUTINE	0.017 AC		70 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.017 AC		92 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		82 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.014 AC		79 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.014 AC		53 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.011 AC	40.2 AC	87 AC	0.5 AC	0.8745 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	147	9471	43278	32	655	2728	4	5	ND	843	56320
06/12/2017	183	12895	43794	39	715	3166	5	5	ND	947	60802

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 6 MONTHS:
THE DETECTION OF HYDROGEN IN CONJUNCTION WITH CARBON OXIDES MIGHT BE AN INDICATIVE OF LOW ENERGY DISCHARGE-CORONA INVOLVING SOLID INSULATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	10	0.82	A
31/07/2017	26	8	1.04	B
30/08/2017	34	6	0.49	A
29/09/2017	36	8	0.57	A
30/10/2017	44	10	0.45	A
27/11/2017	42	8	0.40	A
06/12/2017	38	7	0.46	A

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 1

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.02	0.01	0.01
06/12/2017	ND	0.01	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.140 AC
06/12/2017	0.000 AC	0.040 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.24 AC
06/12/2017	0.27 AC

CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.062mgKOH/g is **QUESTIONABLE** whereas **IFT** of 25.0mN/m is **UNACCEPTABLE**. These two parameters indicate the oil is deteriorated. Remove the unwanted acids which actively promote more acid and oxidation by-product, to avoid more accumulation of contaminants in the oil that will weaken the oil interfacial tension.
- **Dielectric** of 48kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 18ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.89% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furanic compound** 2FOL of 75ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** were not detected, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur test** on copper strip resultant in light orange, thus **Non-corrosive of 1A**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		36		NORMAL		GOOD	
31/07/2017	OFFLINE	28					
30/08/2017	OFFLINE	30				GOOD	NONE
29/09/2017	OFFLINE	35		NORMAL		GOOD	NONE
30/10/2017	OFFLINE	50		NORMAL		GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.056 QU	25.3 UN	50 AC	1.5 AC	0.8529 AC	CLEAR AC	
31/07/2017	ROUTINE	0.059 QU		57 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.056 QU		65 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.059 QU		58 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.059 QU		23 UN	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.062 QU		25 QU	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.062 QU	25.0 UN	48 AC	1.5 AC	0.8583 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	20494	67487	4	69	1143	ND	1	ND	109	89233
06/12/2017	39	19933	67198	4	83	1261	ND	1	ND	127	88519

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	26	18	2.34	D
31/07/2017	25	20	2.60	D
30/08/2017	29	18	1.89	D
29/09/2017	32	19	1.80	D
30/10/2017	42	32	1.60	C
27/11/2017	41	33	1.65	C
06/12/2017	30	18	1.89	D

FIELD SERVICE
DATE SERVICE

No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.053mgKOH/g is **QUESTIONABLE** while **IFT** of 27.3mN/m is **UNACCEPTABLE**. These two parameters indicate there are further accumulation of oxidation byproducts in the oil. Oil oxidation byproducts promotes more acids to be produced by the present acids. Therefore, removal of these contaminations is highly recommended.
- **Dielectric** of 51kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 17ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.78% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furanic compound** 2FOL of 67ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.02% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		37		NORMAL		GOOD	
31/07/2017	OFFLINE						
30/08/2017	OFFLINE	39		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.042 AC	26.6 UN	41 AC	1.5 AC	0.8540 AC	CLEAR AC	
31/07/2017	ROUTINE	0.045 AC		25 QU	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.042 AC		42 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.048 AC		59 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.051 QU		62 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.051 QU		50 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.053 QU	27.3 UN	51 AC	1.5 AC	0.8589 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	27	18639	66781	3	64	1106	ND	1	ND	95	86621
06/12/2017	20	19360	61590	3	45	728	ND	1	ND	69	81747

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	27	21	2.48	D
31/07/2017	24	27	3.91	E
30/08/2017	28	14	1.65	C
29/09/2017	31	19	1.80	D
30/10/2017	35	17	1.21	B
27/11/2017	35	18	1.28	B
06/12/2017	30	17	1.78	D

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	62	ND	ND	ND	62
06/12/2017	ND	67	ND	ND	ND	67

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.900 AC
06/12/2017	0.000 AC	0.980 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.02 UN
06/12/2017	0.02 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.053mgKOH/gm and **IFT** of 28.3mN/m are **QUESTIONABLE**.
These two parameters indicates there are formation of oxidation by products in the oil. Close monitoring is advised.
- **Dielectric** of 57kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 16ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.68% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furanic compound** 2FOL of 14ppb was detected which indicates the solid insulation is wet. It is important to maintain the oil in a good condition so as not to further deteriorate the paper insulation.
- **Metals-In-Oil** were not detected, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.01% by weight is detected, thus **UNACCEPTABLE**.
Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- Retest **Acidity** and **Interfacial Tension** in 6 months to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality.
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		28		NORMAL		GOOD	
31/07/2017	ONLINE						
30/08/2017	OFFLINE	35		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	38		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.048 AC	28.4 QU	44 AC	1.0 AC	0.8537 AC	CLEAR AC	
31/07/2017	ROUTINE	0.042 AC		77 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.042 AC		61 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.045 AC		72 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.048 AC		72 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.051 QU		62 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.053 QU	28.3 QU	57 AC	1.5 AC	0.8624 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	32	15273	65111	5	95	1002	ND	2	ND	134	81520
06/12/2017	7	14730	65738	3	42	617	ND	ND	ND	52	81137

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	25	32	4.16	E
31/07/2017	25	16	2.08	D
30/08/2017	30	13	1.37	C
29/09/2017	32	14	1.33	B
30/10/2017	37	16	1.04	B
27/11/2017	34	17	1.39	C
06/12/2017	30	16	1.68	C

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	32	ND	ND	ND	32
06/12/2017	ND	14	ND	ND	ND	14

RECOMMENDATION:
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.01
06/12/2017	ND	ND	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.700 AC
06/12/2017	0.000 AC	0.630 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.01 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.031mgKOH/g is **ACCEPTABLE** while **IFT** of 27.2mN/m is **UNACCEPTABLE**. The low IFT level indicates the oil is containing contaminants that had weakened the Interfacial tension of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 41kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 18ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.89% M/dw is graded as **D**. The value is approaching the borderline of questionable level. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factor** for 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.06% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017		28		NORMAL		BAD	
31/07/2017	ONLINE						
30/08/2017	OFFLINE	34		NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	44		NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.028 AC	26.3 UN	24 UN	1.5 AC	0.8596 AC	CLEAR AC	
31/07/2017	ROUTINE	0.022 AC		88 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.028 AC		44 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.031 AC		53 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.028 AC		48 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.034 AC		41 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.031 AC	27.2 UN	41 AC	1.5 AC	0.8645 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	13	12265	66770	5	172	1206	ND	8	ND	198	80439
06/12/2017	8	13331	63527	3	70	720	ND	1	ND	82	77660

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	26	29	3.77	E
31/07/2017	26	15	1.95	D
30/08/2017	30	13	1.37	C
29/09/2017	33	15	1.23	B
30/10/2017	39	18	0.99	B
27/11/2017	37	30	1.95	D
06/12/2017	30	18	1.89	D

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG - FUJIAN
DATE MFG - 2010
MAIN TANK S/N - **10034**
KVA -
PRI - 72000 V
SEC - 380 V
KILOGRAMS - 1000
EQUIP TYPE - TRANSFORMER
TRANS CLASS - ONAN
IMPEDENCE -
PHASE/CYCLE - 3/50 Hz
LIQUID TYPE - OIL
LITRES LIQUID - 630

LTC MFG -
LTC S/N -
LTC FAB NO -
LTC MODEL -
LTC LITRES LIQUID -
LTC ENERGIZED -

ADDITIONAL EQUIPMENT

RADIATORS - YES
FANS - NO
H2O COOLED - NO
OIL PUMPS - NO
TOP FPV -
BOTTOM FPV -
BUCHOLZ - YES
CONSERVATOR - YES
LTC COMPARTMENT - NO
BUSHING LOC - TOP
FREE BREATHER - YES
HOSE LENGTH -
POWER AVAILABLE - NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	18	ND	ND	ND	18
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.05	0.02	0.01
06/12/2017	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.180 QU	6.680 UN
06/12/2017	0.010 AC	2.180 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.08 UN
06/12/2017	0.06 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
6.	MAIN TRAFO UNIT 3	1122170	9

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 41.5mN/m are **ACCEPTABLE**.
- **Dielectric** of 84kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- The **DGA** result shows that Carbon Monoxide to have stabilized. Other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 9ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.58% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.22% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE	46	56			GOOD	
31/07/2017	ONLINE	48	50				
30/08/2017	ONLINE	55	60	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE	58	68	NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE	32	32	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE	46	50				
06/12/2017	ONLINE	57	48				

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.017 AC	42.3 AC	78 AC	0.5 AC	0.8780 AC	CLEAR AC	
31/07/2017	ROUTINE	0.014 AC		77 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.017 AC		86 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.014 AC		86 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.014 AC		88 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.014 AC		87 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.014 AC	41.5 AC	84 AC	0.5 AC	0.8832 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	24	10131	44084	11	641	2250	1	1	ND	678	57143
06/12/2017	39	14023	43226	11	668	2456	ND	1	ND	719	60424

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	15	1.23	B
31/07/2017	35	17	1.21	B
30/08/2017	35	8	0.57	A
29/09/2017	42	10	0.50	A
30/10/2017	28	9	1.06	B
27/11/2017	35	8	0.57	A
06/12/2017	37	9	0.58	A

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1122170
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 73920
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.79%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 26708
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.02
06/12/2017	0.01	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.130 AC
06/12/2017	0.000 AC	0.070 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.22 AC

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
7.	UAT 3	1035082	10

A) Analysis

- **Acidity** of 0.020mgKOH/g and **IFT** of 37.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 86kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** reveals further increases in Carbon Monoxide and Dioxide while other gases remain within normal limits. As the amount is significantly high, transformer manufacturer can be consulted for the similar cases with the same design of the transformer. Meanwhile, proceed with annual monitoring.
- **Water Content** in oil of 12ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.66% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.23% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - UAT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		55			BAD	
31/07/2017	ONLINE						
30/08/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE		58	NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE		44	NORMAL	FAIR	GOOD	NONE
27/11/2017	ONLINE		60				
06/12/2017	ONLINE		60				

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	38.1 AC	60 AC	0.5 AC	0.8767 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		88 AC	0.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		89 AC	0.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.020 AC		80 AC	0.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.022 AC		86 AC	0.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.022 AC		86 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.020 AC	37.6 AC	86 AC	0.5 AC	0.8820 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	20	16028	68308	20	971	3536	5	7	ND	1023	88895
06/12/2017	36	18752	66758	19	1116	3622	5	7	ND	1183	90315

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	33	11	0.90	A
31/07/2017	32	14	1.33	B
30/08/2017	35	11	0.78	A
29/09/2017	37	12	0.78	A
30/10/2017	29	11	1.15	A
27/11/2017	37	12	0.78	A
06/12/2017	39	12	0.66	A

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **UAT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- SHANDONG
DATE MFG	- 2010
MAIN TANK S/N	- 1035082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 14600
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 3749
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.02	0.01	0.04
06/12/2017	ND	0.01	0.08

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.390 AC
06/12/2017	0.000 AC	0.170 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.23 AC

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	007308428	11

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 34.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 31kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 31ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.39% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 30ppm. This indicates the presence of high water content in the oil. The presence of high water content could cause dielectric failure.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.18% by weight is detected, thus **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- The Specific Gravity of the oil needs further investigation. Should the unit is safe to operate with the current oil-filled, highly recommended to perform **OIL PURIFICATION**. Result indicates that the moisture content in the transformer oil is not conducive for normal operation of a transformer. Oil purification will remove excessive moisture detected in the transformer oil and thus improving dielectric breakdown voltage of the mineral oil. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
31/07/2017	ONLINE						
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	23.2 UN	32 AC	1.5 AC	0.8432 AC	CLEAR AC	
31/07/2017	ROUTINE	0.028 AC		47 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.028 AC		36 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.025 AC		45 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.025 AC		40 AC	1.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.025 AC		31 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC	34.6 AC	31 AC	1.5 AC	0.8180 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	14	12943	68807	3	62	1243	ND	ND	ND	79	83072
06/12/2017	14	12739	65097	3	66	612	ND	ND	ND	83	78531

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	36	26	1.85	D
31/07/2017	31	21	2.00	D
30/08/2017	30	22	2.31	D
29/09/2017	34	20	1.64	C
30/10/2017	28	19	2.24	D
28/11/2017	43	34	1.53	C
06/12/2017	43	31	1.39	C

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	ND
06/12/2017	0.01	ND	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.350 AC
06/12/2017	0.000 AC	0.040 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.18 QU

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2E	BRASSY GOLD

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.22mgKOH/g is **ACCEPTABLE** while **IFT** of 29.7mN/m is **QUESTIONABLE**. This indicates the presence of oxidation by products in the oil.
- **Dielectric** of 29kV is **QUESTIONABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 44ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.98% M/dw is graded as **D**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm. This indicates the presence of high water content in the oil. The presence of high water content could cause dielectric failure.
- **Furan analysis** indicates the solid insulation is normal. None of furanic compound is detected.
- **Metals-In-Oil** were not detected, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.20% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- The Specific Gravity of the oil needs further investigation. Should the unit is safe to operate with the current oil-filled, highly recommended to perform **OIL Regeneration**. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as excess moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.118 UN	16.2 UN	24 UN	3.0 AC	0.8435 AC	CLEAR AC	
31/07/2017	ROUTINE	0.112 UN		28 QU	3.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.109 UN		34 AC	2.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.115 UN		21 UN	2.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.118 UN		33 AC	2.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.017 AC		25 QU	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.022 AC	29.7 QU	29 QU	0.5 AC	0.8106 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	38	16781	70523	5	129	2431	3	5	ND	180	89915
06/12/2017	42	19523	66059	3	102	912	ND	ND	ND	147	86641

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	37	39	2.54	D
31/07/2017	31	38	3.61	E
30/08/2017	26	17	2.21	D
29/09/2017	36	37	2.63	D
30/10/2017	28	20	2.36	D
28/11/2017	43	28	1.26	B
06/12/2017	43	44	1.98	D

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	169	ND	ND	169
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.330 AC
06/12/2017	0.000 AC	0.030 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.20 AC

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	3A	MAGENTA

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 39.8mN/m are **ACCEPTABLE**.
- **Dielectric** of 45kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 29ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.30% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates the presence of high water content in the oil. The presence of high water content could cause dielectric failure.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** were not detected, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.21% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- The Specific Gravity of the oil needs further investigation. Should the unit is safe to operate with the current oil-filled, highly recommended to perform **OIL PURIFICATION**. Result indicates that the moisture content in the transformer oil is not conducive for normal operation of a transformer. Oil purification will remove excessive moisture detected in the transformer oil and thus improving dielectric breakdown voltage of the mineral oil. Furthermore, passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE						
31/07/2017	ONLINE						
30/08/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	ONLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			LOW	FAIR	GOOD	NONE
28/11/2017	ONLINE						
06/12/2017	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.025 AC	27.8 UN	40 AC	1.5 AC	0.8415 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		66 AC	1.5 AC		CLEAR AC	
30/08/2017	ROUTINE	0.025 AC		50 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.028 AC		59 AC	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.028 AC		55 AC	1.5 AC		CLEAR AC	
28/11/2017	ROUTINE	0.014 AC		39 AC	0.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.014 AC	39.8 AC	45 AC	0.5 AC	0.8099 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	21	16364	70694	3	38	1068	ND	ND	ND	62	88188
06/12/2017	15	14515	65277	2	40	559	ND	ND	ND	57	80408

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	37	27	1.75	D
31/07/2017	29	18	1.89	D
30/08/2017	35	21	1.49	C
29/09/2017	37	22	1.43	C
30/10/2017	28	16	1.89	D
28/11/2017	44	24	1.08	B
06/12/2017	44	29	1.30	B

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.620 AC
06/12/2017	0.000 AC	0.080 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.21 AC

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2E	BRASSY GOLD

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.25mgKOH/g is **ACCEPTABLE** while **IFT** of 29.8mN/m is **QUESTIONABLE**. This indicates the presence of oxidation by products in the oil.
- **Dielectric** of 43kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 19ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.80% M/dw is graded as **D**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.03% by weight is detected, thus **UNACCEPTABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

B) Recommendation

- Retest **Acidity** and **Interfacial Tension** in 6 months to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
30/08/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
29/09/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
30/10/2017	OFFLINE			NORMAL	FAIR	GOOD	NONE
28/11/2017	OFFLINE						
06/12/2017	OFFLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.020 AC	31.2 QU	62 AC	1.0 AC	0.8420 AC	CLEAR AC	
04/07/2017	ROUTINE	0.022 AC		60 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.020 AC		55 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		59 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.025 AC		55 AC	1.0 AC		CLEAR AC	
28/11/2017	ROUTINE	0.022 AC		69 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.025 AC	29.8 QU	43 AC	1.0 AC	0.8471 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	10	18669	66036	3	24	665	ND	ND	ND	37	85407
06/12/2017	14	15414	63712	2	22	675	ND	ND	ND	38	79839

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	32	19	1.80	D
04/07/2017	31	15	1.42	C
30/08/2017	27	17	2.01	D
29/09/2017	29	15	1.58	C
30/10/2017	28	15	1.77	D
28/11/2017	29	16	1.68	C
06/12/2017	32	19	1.80	D

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	ND	ND
06/12/2017	ND	0.03	ND

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.360 AC
06/12/2017	0.000 AC	0.370 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.03 UN

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1B	DARK ORANGE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
12.	ASH HANDLING	10820108	15

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 37.5mN/m are **ACCEPTABLE**.
- **Dielectric** of 34kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- The **DGA** result shows that Carbon Monoxide and Carbon Dioxide level are high. Other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 26ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.85% M/dw is graded as **B**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.24% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in multicolored with lavender blue and/or silver overlaid on claret red, thus **Non-corrosive of 2C**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE	46					
31/07/2017	ONLINE						
30/08/2017	OFFLINE	30		NORMAL	FAIR		NONE
29/09/2017	OFFLINE	40		NORMAL	FAIR		NONE
30/10/2017	ONLINE	45		NORMAL	FAIR		NONE
27/11/2017	ONLINE	46					
06/12/2017	ONLINE	52					

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	38.1 AC	38 AC	1.0 AC	0.8779 AC	CLEAR AC	
31/07/2017	ROUTINE	0.025 AC		56 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		53 AC	1.0 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		57 AC	1.0 AC		CLEAR AC	
30/10/2017	ROUTINE	0.022 AC		45 AC	1.0 AC		CLEAR AC	
27/11/2017	ROUTINE	0.020 AC		36 AC	1.0 AC		CLEAR AC	
06/12/2017	ROUTINE	0.017 AC	37.5 AC	34 AC	1.0 AC	0.8832 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	10984	71058	20	441	5037	7	8	ND	511	87590
06/12/2017	49	12034	67124	20	558	5440	8	8	ND	643	85241

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	34	28	2.30	C
31/07/2017	26	21	2.73	D
30/08/2017	28	16	1.89	B
29/09/2017	34	16	1.31	B
30/10/2017	33	20	1.64	B
27/11/2017	36	29	2.06	C
06/12/2017	36	26	1.85	B

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.730 AC
06/12/2017	0.000 AC	0.240 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.24 AC

®CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE

®PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
13.	ASH HANDLING	10820107	16

A) Analysis

- **Acidity** of 0.020mgKOH/g and **IFT** of 36.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 24kV is **UNACCEPTABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** shows significant amount of Carbon Dioxide while other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 32ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.62% M/dw is graded as **D**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm. This indicates the presence of high water content in the oil. The presence of high water content could cause dielectric failure.
- **Furanic compound** 2FAL of 3ppb was detected which indicates normal aging of the solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.24% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in light orange, thus **Non-corrosive of 1A**.

B) Recommendation

- Highly recommended to perform **OIL PURIFICATION**. Result indicates that the moisture content in the transformer oil is not conducive for normal operation of a transformer. Oil purification will remove excessive moisture detected in the transformer oil and thus improving dielectric breakdown voltage of the mineral oil. Furthermore, passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

DATE PRINTED 23/12/2017
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
04/07/2017	ONLINE		40				
31/07/2017	ONLINE						
30/08/2017	ONLINE	47		NORMAL	FAIR		NONE
29/09/2017	ONLINE	60		NORMAL	FAIR		NONE
30/10/2017	OFFLINE	32		NORMAL	FAIR		NONE
27/11/2017	OFFLINE	40					
06/12/2017	ONLINE	48					

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1082017
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
04/07/2017	ROUTINE	0.022 AC	37.3 AC	36 AC	1.0 AC	0.8788 AC	CLEAR AC	
31/07/2017	ROUTINE	0.022 AC		40 AC	1.0 AC		CLEAR AC	
30/08/2017	ROUTINE	0.022 AC		31 AC	1.5 AC		CLEAR AC	
29/09/2017	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	
30/10/2017	ROUTINE	0.020 AC		46 AC	1.5 AC		CLEAR AC	
27/11/2017	ROUTINE	0.022 AC		47 AC	1.5 AC		CLEAR AC	
06/12/2017	ROUTINE	0.020 AC	36.6 AC	24 UN	1.5 AC	0.8840 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	18	21157	64224	4	129	1676	ND	1	ND	152	87209
06/12/2017	20	15791	63530	5	275	2783	ND	4	ND	304	82408

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON DIOXIDE GAS LEVEL IS HIGH.OTHER GASES ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
04/07/2017	29	21	2.20	C
31/07/2017	31	32	3.04	D
30/08/2017	34	29	2.38	C
29/09/2017	40	31	1.71	B
30/10/2017	29	21	2.20	C
27/11/2017	29	20	2.10	C
06/12/2017	34	32	2.62	D

FIELD SERVICE
DATE SERVICE

DATE PRINTED 23/12/2017

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - ASH HANDLING

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1082017
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	3	ND	ND	3

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	0.01	0.01

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.350 AC
06/12/2017	0.000 AC	0.270 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.24 AC

CORROSIVE SULFUR (ASTM 1275B)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1A	LIGHT ORANGE

PASSIVATOR COUNT, EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

3.0 Summary of Our Comments & Recommendations:

No.	Sub Name	Unit No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	1006901	4	- Retest 6 months: DGA - Add passivator - Other parameters: Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	Perform OIL REGENERATION
3.	TRAFO 2 ESP UNIT 1	10035	6	Perform OIL REGENERATION
4.	TRAFO 3 ESP UNIT 1	10031	7	- Retest 6 months: Acidity and IFT - Add inhibitor 2,6-DBPC and passivator - Other parameters: Periodic condition monitoring
5.	TRAFO 4 ESP UNIT 1	10034	8	Perform OIL REGENERATION
6.	MAIN TRAFO UNIT 3	1122170	9	- Add passivator - Periodic condition monitoring
7.	UAT 3	1035082	10	- Add passivator - Periodic condition monitoring
8.	TRAFO 1 ESP UNIT 3	007308428	11	Perform OIL PURIFICATION
9.	TRAFO 2 ESP UNIT 3	0070084208	12	Perform OIL REGENERATION
10.	TRAFO 3 ESP UNIT 3	0072084208	13	Perform OIL PURIFICATION
11.	TRAFO 4 UNIT 3	0071084208	14	- Retest 6 months: Acidity and IFT - Add inhibitor 2,6-DBPC and passivator - Other parameters: Periodic condition monitoring
12.	ASH HANDLING	10820108	15	- Add passivator - Periodic condition monitoring
13.	ASH HANDLING	10820107	16	Perform OIL PURIFICATION

Page Disclaimer :

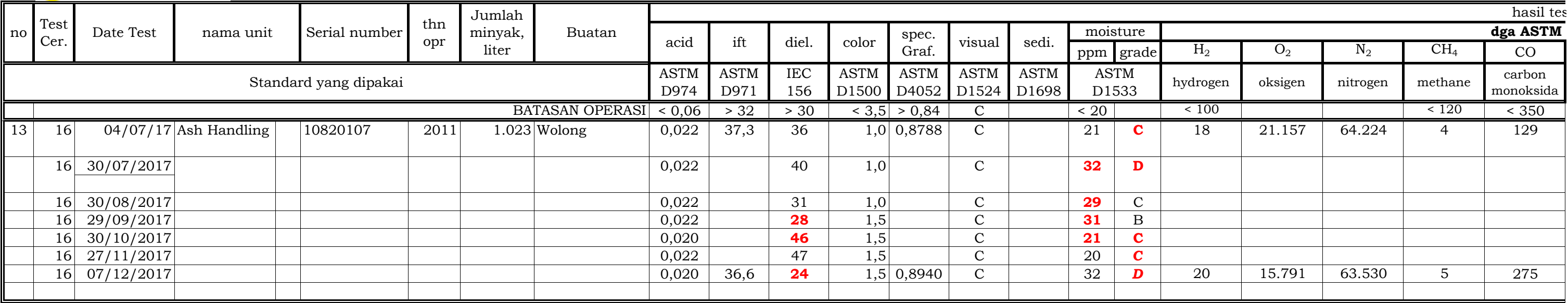
All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

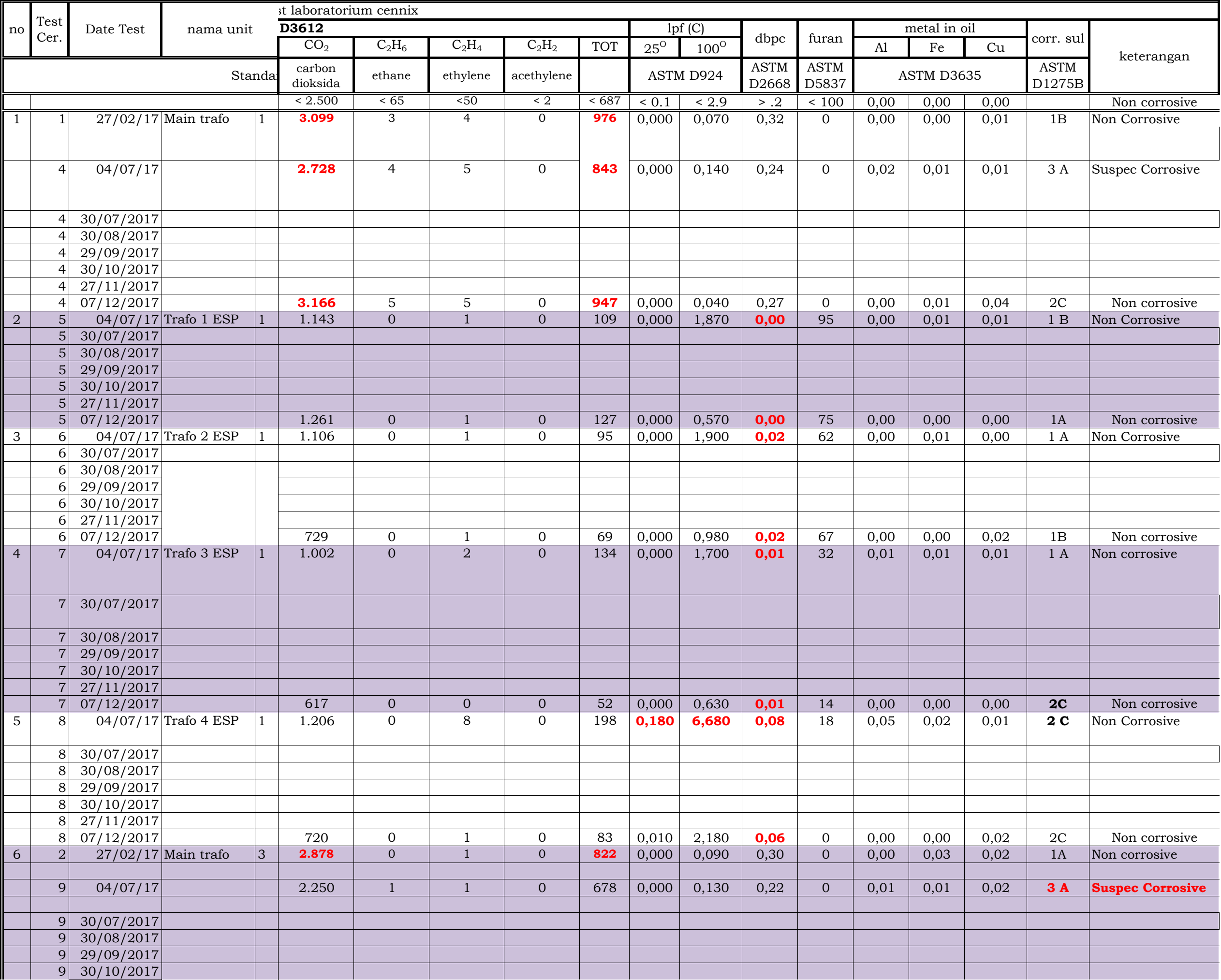


no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil tes														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida		
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	
1	1	27/02/17	Main trafo	1	1006901	2011	30.857	Wolong	0,014	39,9	99	0,5	0,8693	C		9	A	135	15.834	57.625	27	807
	4	04/07/17							0,017	41,5	58	0,5	0,8691	C		10	A	147	9.471	4.328	32	655
	4	30/07/2017							0,017		70	0,5		C		8	B					
	4	30/08/2017							0,017		92	0,5		C		6	A					
	4	29/09/2017							0,014		82	0,5		C		8	A					
	4	30/10/2017							0,014		82	0,5		C		6	A					
	4	27/11/2017							0,014		53	0,5		C		8	A					
	4	07/12/2017							0,011	40,1	87	0,5	0,8745	C		7	A	183	12.895	43.794	39	715
2	5	04/07/17	Trafo 1 ESP	1	10937	2010	1.143	Fujia	0,056	25,3	50	1,5	0,8529	C		18	D	35	20.494	67.487	4	69
	5	30/07/2017							0,059		57	1,5		C		20	D					
	5	30/08/2017							0,056		65	1,5		C		18	D					
	5	29/09/2017							0,059		58	1,5		C		19	D					
	5	30/10/2017							0,059		23	1,5		C		32	C					
	5	27/11/2017							0,062		25	1,5		C		33	C					
	5	07/12/2017								25,0	49	1,5	0,8593	C		18	D	39	19.933	67.198	4	83
3	6	04/07/17	Trafo 2 ESP	1	10035	2010	1.143	Fujia	0,042	26,6	41	1,5	0,8540	C		21	D	27	18.639	66.781	3	64
	6	30/07/2017							0,045		25	1,5		C		27	E					
	6	30/08/2017							0,042		42	1,5		C		14	C					
	6	29/09/2017							0,048		59	1,5		C		19	D					
	6	30/10/2017							0,051		62	1,5		C		17	B					
	6	27/11/2017							0,051		50	1,5		C		18	B					
	6	07/12/2017							0,063	27,3	51	1,5	0,8589	C		17	D	20	19.360	61.590	3	45
4	7	04/07/17	Trafo 3 ESP	1	10031	2010	1.143	Fujia	0,048	28,4	44	1,0	0,8537	C		32	E	32	15.273	65.111	5	95
	7	30/07/2017							0,042		77	1,5				10	D					
	7	30/08/2017							0,042		61	1,5		C		13	C					
	7	29/09/2017							0,045		72	1,5		C		14	B					
	7	30/10/2017							0,048		72	1,5		C		16	B					
	7	27/11/2017							0,051		62	1,5		C		17	C					
	7	07/12/2017							0,063	29,3	57	1,5	0,8624	C		16	C	7	14.730	65.738	3	42
5	8	04/07/17	Trafo 4 ESP	1	10034	2010	1.143	Fujia	0,028	26,3	24	1,5	0,8596	C		29	E	13	12.265	66.770	5	172
	8	30/07/2017							0,022		88	1,5		C		15	D					
	8	30/08/2017							0,028		44	1,5		C		13	C					
	8	29/09/2017							0,031		53	1,0		C		15	B					
	8	30/10/2017							0,028		48	1,0		C		18	B					
	8	27/11/2017							0,034		41	1,5		C		30	D					
	8	07/12/2017							0,031	27,2	41	1,5	0,8645	C		18	D	9	13.331	63.527	3	70
6	2	27/02/17	Main trafo	3	1122170	2010	26.706	Shandong	0,017	39,7	87	0,5	0,8781	C		9	A	4	13.330	43.140	9	808
	9	04/07/17							0,017	42,3	78	0,5	0,8780	C		15	B	24	10.131	44.084	11	641
	9	30/07/2017							0,014		77	0,5		C		17	B					
	9	30/08/2017							0,017		86	0,5		C		8	A					
	9	29/09/2017							0,014		86	0,5		C		10	A					
	9	30/10/2017							0,014		88	0,5		C		9	A					



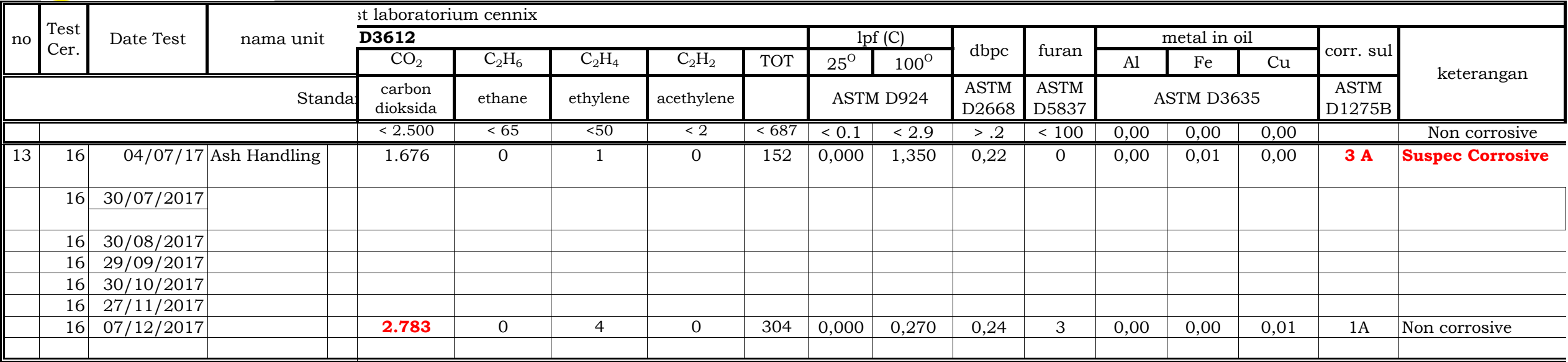
no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil tes													
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM				
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350
	9	27/11/2017						0,014		87	0,5		C		8	A					
	9	07/12/2017						0,014	41,5	84	0,5	0,8932	C			A	39	14.023	43.226	11	668
7	3	27/02/17	Trafo UAT	3	1035082	2010	4.749 Shandong	0,017	37,6	94	0,5	0,8768	C		9	A	20	12.535	56.879	20	985
	10	04/07/17						0,022	38,1	60	0,5	0,8767	C		11	A	20	16.028	68.308	20	971
	10	30/07/2017						0,025		88	0,5		C		14	B					
	10	30/08/2017						0,022		89	0,5		C		11	A					
	10	29/09/2017						0,020		80	0,5		C		12	A					
	10	30/10/2017						0,022		86	0,5		C		11	A					
	10	27/11/2017						0,022		86	0,5		C		12	A					
	10	07/12/2017						0,020	37,6	86	0,5	0,8200	C		12	A	36	18.752	66.758	1	1.116
8	11	04/07/17	Trafo 1 ESP	3	007308428	2010	731 Zhejian	0,022	23,2	32	1,5	0,8432	C		26	D	14	12.943	68.807	3	62
	11	30/07/2017						0,028		47	1,5		C		21	D					
	11	30/08/2017						0,028		36	1,5		C		22	D					
	11	29/09/2017						0,025		45	1,5		C		20	C					
	11	30/10/2017						0,025		40	1,5		C		19	D					
	11	27/11/2017						0,025		31	1,5		C		34	C					
	11	07/12/2017						0,022	34,6	31	1,5	0,8190	C		31	C	14	12.739	65.097	3	66
9	12	04/07/17	Trafo 2 ESP	3	007008428	2010	731 Zhejian	0,118	16,2	24	3,0	0,8435	C		39	D	38	16.781	70.523	5	129
	12	30/07/2017						0,112		28	3,0		C		38	E					
	12	30/08/2017						0,109		34	2,5		C		17	D					
	12	29/09/2017						0,115		21	2,5		C		37	D					
	12	30/10/2017						0,118		33	2,5		C		20	D					
	12	27/11/2017						0,017		25	0,5		C		28	B					
	12	07/12/2017						0,022	29,7	29	0,1	0,8106	C		44	D	42	19.523	66.059	3	102
10	13	04/07/17	Trafo 3 ESP	3	007208428	2010	731 Zhejian	0,025	27,8	40	1,5	0,8415	C		27	D	21	16.364	70.694	3	38
	13	30/07/2017						0,025		66	1,5		C		18	D					
	13	30/08/2017						0,025		50	1,5		C		21	C					
	13	29/09/2017						0,028		59	1,5		C		22	C					
	13	30/10/2017						0,028		55	1,5		C		28	D					
	13	27/11/2017						0,014		39	0,5		C		24	B					
	13	07/12/2017						0,014	39,8	45	0,5	0,8099	C		29	B	15	14.515	65.277	2	40
11	14	04/07/17	Trafo 4 ESP	3	007108428	2010	731 Zhejian	0,020	31,2	62	1,0	0,8420	C		19	D	10	18.669	66.036	3	24
	14	30/07/2017						0,022		60	1,0		C		15	C					
	14	30/08/2017						0,020		55	1,0		C		17	D					
	14	29/09/2017						0,022		59	1,0		C		15	C					
	14	30/10/2017						0,025		55	1,0		C		15	D					
	14	27/11/2017						0,022		69	1,0		C		16	C					
	14	07/12/2017						0,025	29,8	43	1,0	0,8471	C		19	D	14	15.414	63.712	2	22
12	15	04/07/17	Ash Handling		10820108	2011	1.023 Wolong	0,022	38,1	38	1,0	0,8779	C		28	C	35	10.984	71.058	20	441
	15	30/07/2017						0,025		56	1,0		C		21	D					
	15	30/08/2017						0,022		53	1,0		C		16	B					
	15	29/09/2017						0,022		57	1,0		C		16	B					
	15	30/10/2017						0,022		45	1,0		C		20	B					
	15	27/11/2017						0,020		36	1,0		C		29	C					
	15	07/12/2017						0,017	37,5	34	1,0	0,8932	C		26	B	49	12.034	67.124	20	558







no	Test Cer.	Date Test	nama unit	st laboratorium cennix														
				D3612						lpf (C)		dbpc	furan	metal in oil			corr. sul	keterangan
					CO ₂	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu		
					carbon dioksida	ethane	ethylene	acethylene		ASTM D924				ASTM D2668	ASTM D5837	ASTM D3635		
					< 2.500	< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive
	9	27/11/2017																
	9	07/12/2017			2.456	0	1	0	719	0,000	0,070	0,22	0	0,01	0,00	0,03	2C	Non corrosive
7	3	27/02/17	Trafo UAT	3	3.877	6	7	0	1.038	0,000	0,100	0,30	0	0,00	0,01	0,04	1A	Non Corrosive
	10	04/07/17			3.536	5	7	0	1.023	0,000	0,390	0,23	0	0,02	0,01	0,04	3 A	Suspec Corrosive
	10	30/07/2017																
	10	30/08/2017																
	10	29/09/2017																
	10	30/10/2017																
	10	27/11/2017																
	10	07/12/2017			3.622	5	7	0	1.165	0,000	0,170	0,23	0	0,00	0,01	0,08	2C	Non corrosive
8	11	04/07/17	Trafo 1 ESP	3	1.243	0	0	0	79	0,000	0,350	0,00	0	0,01	0,01	0,00	1 B	Non corrosive
	11	30/07/2017																
	11	30/08/2017																
	11	29/09/2017																
	11	30/10/2017																
	11	27/11/2017																
	11	07/12/2017			612	0	0	0	83	0,000	0,040	0,18	0	.01	0,00	0,00	2E	Non corrosive
9	12	04/07/17	Trafo 2 ESP	3	2.431	3	5	0	180	0,000	1,330	0,00	169	0,00	0,01	0,01	2 C	Non corrosive
	12	30/07/2017																
	12	30/08/2017																
	12	29/09/2017																
	12	30/10/2017																
	12	27/11/2017																
	12	07/12/2017			912	0	0	0	147	0,000	0,030	0,20	0	0,00	0,00	0,00	3A	Non corrosive
10	13	04/07/17	Trafo 3 ESP	3	1.068	0	0	0	62	0,000	0,620	0,01	0	0,00	0,01	0,00	2 C	Non corrosive
	13	30/07/2017																
	13	30/08/2017																
	13	29/09/2017																
	13	30/10/2017																
	13	27/11/2017																
	13	07/12/2017			559	0	0	0	57	0,000	.08	.21	0	0,00	0,00	0,00	2E	Non corrosive
11	14	04/07/17	Trafo 4 ESP	3	665	0	0	0	37	0,000	1,360	0,01	0	0,00	0,00	0,00	3 A	Suspec Corrosive
	14	30/07/2017																
	14	30/08/2017																
	14	29/09/2017																
	14	30/10/2017																
	14	27/11/2017																
	14	07/12/2017			675	0	0	0	38	0,000	0,370	0,03	0	0,00	0,00	0,00	1B	Non corrosive
12	15	04/07/17	Ash Handling		5.037	7	8	0	511	0,000	0,730	0,23	0	0,00	0,01	0,01	1 B	Non corrosive
	15	30/07/2017																
	15	30/08/2017																
	15	29/09/2017																
	15	30/10/2017																
	15	27/11/2017																
	15	07/12/2017			5.440	8	9	0	644	0,000	0,240	0,24	0	0,00	0,00	0,01	2C	Non corrosive





no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
1	1	27/02/17	Main trafo	1	Retest 6 months: DGA Check leakages Other parameters: Periodic Condition Monitoring	FOR TEST 07 DECEMBER 2017 Untuk Main Trafo 1 dan 3 serta Trafo UAT 3 yang tinggi CO dan CO2 nya, harap diperhatikan masalah cooling system antara lain menyalakan fan sejak awal trafo dinyalakan. Salah satu akibat dari CO dan CO2 yang tinggi (yang artinya sedang mengalami gangguan thermal secara internal) Mengingat banyak trafo ESP (Tc 5,6,7,8,11,12,13) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami saat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg diatas 7 cycle Trafo ESP (Tc 6,7,8) Furan (FOL) banyak berkurang karena moisture cukup banyak dalam kertas dan kemungkinan juga kualitas kertas kurang bagus Untuk trafo Ash Handling sebaiknya segera dipurifikasi juga dengan temperatur diatas 650c dan vacuum diatas 74 CMHg sekitar >7 cycle FOR TEST 27 NOVEMBER 2017 Keadaan masih relatif tidak berubah dibandingkan dengan hasil test pada bulan yang sebelumnya. Jadi rekomendasi saya masih sama seperti yang bulan lalu, FOR TEST 30 OKTOBER 2017 Keadaan masih relatif tidak berubah dibandingkan dengan hasil test pada bulan yang sebelumnya. Jadi rekomendasi saya masih sama seperti yang bulan lalu, FOR TEST 29 SEPTEMBER 2017 Yang perlu diganti minyak hanyalah Tc 12. Karena minyak sedikit, jauh lebih murah dibanding regenerasi. Yang perlu dipurifikasi adalah Tc 5, 6,11, 14, 16. Mengingat banyak trafo ESP (Tc 5,6,7,8,11,12,13,14) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami sat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg pelaksanaan diatas 7 cycle. Dari hasil terakhir nampak perubahan moisture pada isolasi
	4	04/07/17			Retest 6 months: DGA Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	4	30/07/2017			Periodic condition monitoring	
	4	30/08/2017			Periodic condition monitoring	
	4	29/09/2017			Periodic condition monitoring	
	4	30/10/2017			Periodic condition monitoring	
	4	27/11/2017			Periodic condition monitoring	
	4	07/12/2017			Retest 6 mnth : DGA	
2	5	04/07/17	Trafo 1 ESP	1	Perform OIL REGENERATION	
	5	30/07/2017			Perform OIL REGENERATION	
	5	30/08/2017			Perform OIL REGENERATION	
	5	29/09/2017			Perform OIL REGENERATION	
	5	30/10/2017			Perform OIL REGENERATION	
	5	27/11/2017			Perform OIL REGENERATION	
	5	07/12/2017			Perform OIL REGENERATION	
3	6	04/07/17	Trafo 2 ESP	1	Perform OIL REGENERATION	
	6	30/07/2017			Perform OIL REGENERATION	
	6	30/08/2017			Perform OIL REGENERATION	
	6	29/09/2017			Perform OIL REGENERATION	
	6	30/10/2017			Perform OIL REGENERATION	
	6	27/11/2017			Perform OIL REGENERATION	
	6	07/12/2017			Perform OIL REGENERATION	
4	7	04/07/17	Trafo 3 ESP	1	Retest 6 months: Acidity, IFT, LPF, Diel and Water Content Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	7	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	7	30/08/2017			Periodic condition monitoring	
	7	29/09/2017			Periodic condition monitoring	
	7	30/10/2017			Periodic condition monitoring	
	7	27/11/2017			Periodic condition monitoring	
	7	07/12/2017			Retest 6 months: Acidity, IFT	
5	8	04/07/17	Trafo 4 ESP	1	Perform OIL REGENERATION Inspect silica gel	
	8	30/07/2017			Perform OIL REGENERATION	
	8	30/08/2017			Perform OIL REGENERATION	
	8	29/09/2017			Perform OIL REGENERATION	
	8	30/10/2017			Perform OIL REGENERATION	
	8	27/11/2017			Perform OIL REGENERATION	
	8	07/12/2017			Perform OIL REGENERATION	
6	2	27/02/17	Main trafo	3	Retest 6 months: DGA Other parameters: Periodic Condition Monitoring	
	9	04/07/17			Add Passivator Periodic Condition Monitoring	
	9	30/07/2017			Periodic condition monitoring	
	9	30/08/2017			Periodic condition monitoring	
	9	29/09/2017			Periodic condition monitoring	
	9	30/10/2017			Periodic condition monitoring	



no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar						
	9	27/11/2017			Periodic condition monitoring	("Grade") pada trafo2 ESP naik turun ini menandakan isolasi sudah mulai kurang bagus. FOR TEST 30 AUGUST 2017 Meningat banyak trafo ESP (Tc 5,6,7,8,11,12,13,14) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja. Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami sat penggantian minyak melalui purifikasi dengan temperatur lebih dari 650C dan vacuum diatas 74 CMHg pelaksanaan diatas 7 cycle Dari hasil terakhir nampak perubahan moisture pada isoilasi ("Grade") pada trafo2 ESP naik turun ini menandakan isolasi kurang bagus. FOR TEST 30 JULY 2017 Untuk trafo ESP 2 Unit 3 sebaiknya minyaknya segera diganti saja, karena Acidity dan Grade Water Content sudah sangat di luar standard Untuk trafo Ash Handling , segera dipurifikasi karena water content sudah jauh di atas standard. FOR TEST 04 JULY 2017 Untuk Main Trafo 1 dan 3 serta Trafo UAT 3 yang tinggi CO dan CO ₂ nya, harap diperhatikan masalah cooling system antara lain menyalakan fan sejak awal trafo dinyalakan. Salah satu akibat dari CO dan CO ₂ yang tinggi (yang artinya sedang mengalami gangguan thermal secara internal) adalah penurunan kondisi Corrosive Sulphur. Dari data yang ada, pada bulan February yang lalu, masih "Non Corrosive", dan pada bulan Juli ini menjadi "Suspec Corrosive". Meningat banyak trafo ESP (Tc 5,6,7,8,11,12,13) yang kurang memenuhi syarat. Kami sarankan untuk diganti minyak saja . Selain jumlah minyaknya sedikit biaya juga akan jauh lebih murah, Saran kami saat penggantian minyak melalui purifikasi dengan temperatur lebih dari 65 ⁰ C dan vacuum diatas 74 CMHg diatas 7 cycle Trafo ESP (Tc 6,7,8) Furan (FOL) banyak berkurang karena moisture cukup banyak dalam kertas dan kemungkinan juga kualitas kertas kurang bagus Untuk trafo Ash Handling sebaiknya segera dipurifikasi juga dengan temperatur diats 65 ⁰ c dan vacuum diatas 74 CMHg sekitar >7 cycle
	9	07/12/2017			Periodic condition monitoring	
7	3	27/02/17	Trafo UAT	3	Retest 6 months: DGA Other parameters: Periodic Condition Monitoring	
	10	04/07/17			Add Passivator Periodic Condition Monitoring Inspect silica gel	
	10	30/07/2017			Periodic condition monitoring	
	10	30/08/2017			Periodic condition monitoring	
	10	29/09/2017			Periodic condition monitoring	
	10	30/10/2017			Periodic condition monitoring	
	10	27/11/2017			Periodic condition monitoring	
	10	07/12/2017			Periodic condition monitoring	
8	11	04/07/17	Trafo 1 ESP	3	Perform OIL REGENERATION	
	11	30/07/2017			Perform OIL REGENERATION	
	11	30/08/2017			Perform OIL REGENERATION	
	11	29/09/2017			Perform OIL REGENERATION	
	11	30/10/2017			Perform OIL REGENERATION	
	11	27/11/2017			Perform OIL REGENERATION	
	11	07/12/2017			Perform OIL PURIFICATION	
9	12	04/07/17	Trafo 2 ESP	3	Perform OIL REGENERATION	
	12	30/07/2017			Perform OIL REGENERATION	
	12	30/08/2017			Perform OIL REGENERATION	
	12	29/09/2017			Perform OIL REGENERATION	
	12	30/10/2017			Perform OIL REGENERATION	
	12	27/11/2017			Perform OIL REGENERATION	
	12	07/12/2017			Perform OIL REGENERATION	
10	13	04/07/17	Trafo 3 ESP	3	Perform OIL REGENERATION	
	13	30/07/2017			Perform OIL REGENERATION	
	13	30/08/2017			Perform OIL REGENERATION	
	13	29/09/2017			Perform OIL REGENERATION	
	13	30/10/2017			Perform OIL REGENERATION	
	13	27/11/2017			Perform OIL REGENERATION	
	13	07/12/2017			Perform OIL PURIFICATION	
11	14	04/07/17	Trafo 4 ESP	3	Retest 6 months: Acidity, IFT and LPF Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring	
	14	30/07/2017			Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring	
	14	30/08/2017			Periodic condition monitoring	
	14	29/09/2017			Periodic condition monitoring	
	14	30/10/2017			Periodic condition monitoring	
	14	27/11/2017			Periodic condition monitoring	
	14	07/12/2017			Retest 6 months: Acidity, IFT	
12	15	04/07/17	Ash Handling		Add Passivator Periodic Condition Monitoring	
	15	30/07/2017			Periodic condition monitoring	
	15	30/08/2017			Periodic condition monitoring	
	15	29/09/2017			Periodic condition monitoring	
	15	30/10/2017			Periodic condition monitoring	
	15	27/11/2017			Periodic condition monitoring	
	15	07/12/2017			Periodic condition monitoring	



no	Test Cer.	Date Test	nama unit	MALAYSIAN LAB RECOMMENDATION	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar					
13	16	04/07/17	Ash Handling	Add Passivator Periodic Condition Monitoring	
	16	30/07/2017		Retest 6 months: Dielectric and Water Content Other parameters: Periodic Condition Monitoring	
	16	30/08/2017		Periodic Condition Monitoring	
	16	29/09/2017		Retest 6 months: Dielectric and Water Content	
	16	30/10/2017		Retest 6 months: Dielectric and Water Content	
	16	27/11/2017		Periodic Condition Monitoring	
	16	07/12/2017		Perform OIL PURIFICATION	

Summary by Soetjahjono
TxM Indonesia